ABHISHEK ZIRANGE

AI/ML & Product Engineer | Agentic AI | Multi-Agent Systems | Enterprise Analytics Platforms

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CAREER SNAPSHOT

✓ 8 + Years Building Enterprise AI & Analytics Systems
✓ Current: Agentic MMM Platform (Fortune 50)
✓ End-to-End MMM Modeling & Automation
✓ Records Deduplicated & Processed Weekly
✓ Multi-Agent LangGraph & Databricks Architecture
✓ AI Delivery Across Retail, Telecom & FMCG Domains

CORE COMPETENCIES

- Agentic AI & LLM Orchestration
- Marketing Mix Modeling (MMM)
- Multi-Agent Systems (LangGraph)
- Data Engineering & Databricks
- Computer Vision & Deep Learning
- Document Intelligence & OCR
- Telecom Analytics
- Full-Stack AI Platforms

PROFESSIONAL SUMMARY

AI/ML and Product Engineer with 8+ years of experience building scalable AI-driven products, agentic AI systems, and marketing analytics platforms across retail, telecom, and FMCG domains. Specialized in Marketing Mix Modeling (MMM), LangGraph-based AI orchestration, NLP, computer vision, and cloud-native ML systems. Proven expertise architecting end-to-end AI solutions using Python, PySpark, Databricks, React, Node.js, and cloud platforms including AWS, GCP, and Azure. Currently architecting an agentic AI-driven Marketing Mix Modeling platform for a Fortune 50 global retailer. Experienced leading cross-functional teams, developing production-grade AI pipelines, and delivering enterprise-scale analytics platforms for Fortune 50 clients.

TECHNICAL SKILLS

Agentic AI & Full-Stack	LangGraph, LangChain, OpenAI API, ChromaDB (Vector DB), React.js, Node.js, Vite, FastAPI, Prompt Engineering, Structured Outputs, Workflow Orchestration
Data & Cloud Engineering	SQL, PySpark, Databricks, DBFS, Spark, CSV-to-Parquet Optimization, Scalable ETL Pipelines, AWS, GCP, Azure, Docker, CI/CD, MLflow Monitoring
Machine Learning	OLS/Ridge Regression, XGBoost, Random Forest, Decision Trees, K-Means Clustering, Linear Regression, Fuzzy Matching, Classification Models, Feature Engineering, Anomaly Detection, Rule-Based Algorithms
Computer Vision & Deep Learning	CNNs, ResNet-50, YOLOv4, Siamese Networks, OCR, OpenCV, Keras, PyTorch, Document & Image Classification, MultiOutputRegressor
NLP & Core Libraries	NLP Pipelines, Python, Pandas, NumPy, SciPy, scikit-learn, Celery
Databases & Visualization	ChromaDB, PostgreSQL, SQLite, Hive, Plotly, Matplotlib, Tableau
Production & Version Control	REST APIs, JSON, Django, Git, GitHub, Docker, VS Code

PROFESSIONAL EXPERIENCE

TENARAI

Jul 2022 – Present

End-to-End MMM Platform with Agentic AI – Fortune 50 Global Retailer

Jan 2025 – Present

Role: Consultant – Analytics

Databricks • OpenAI • LangGraph • Python • PySpark • OLS/Ridge Regression • Parquet • MLflow • React.js • Node.js • REST APIs

- Built an end-to-end, agentic AI-driven MMM platform for a Fortune 50 global retailer covering data ingestion, feature engineering, model training, validation, results, and a business-facing web app.
- Prepared weekly marketing spend, sales, promotion, and external data (holidays, events, seasonality) using PySpark and Databricks pipelines; moved workflows from CSV to Parquet, cutting processing time by more than 40%.
- Built OLS and Ridge regression models to estimate channel contribution, base vs. incremental sales, and marketing ROI, using Ridge to handle correlated media variables.
- Migrated the modeling ecosystem to LangGraph-based agents on Databricks to automate hyperparameter tuning, model execution, and evaluation, with each iteration guided by model results.
- Designed the agent decision loop: after each model run, the agent compares metrics against thresholds and prior runs, then adjusts parameters and re-runs, accepts the model, or escalates for human review.
- Implemented automated guardrails, validation checks, and explainability so model outputs are reliable, transparent, and aligned with business expectations.
- Added pre-modeling data-quality gates (schema, null rates, volume) and post-modeling checks (coefficient stability, prediction shift vs. historical baseline) before results reach users.
- Built explainability views showing which channels and control variables drive results, so non-technical stakeholders can review budget reallocation recommendations with confidence.
- Built an anomaly detection framework to catch data quality issues, report inconsistencies, and model drift early, reducing reporting errors by over 25%.
- Built a brand deduplication system using fuzzy matching, classification models, and API enrichment to clean records.
- Directed engineering efforts and worked with cross-functional stakeholders on product strategy, architecture, and roadmap, evolving the POC into an enterprise MMM product.

Telecom Analytics & Subscriber Intelligence Solutions — South Africa & Nigeria

Dec 2022 – Jan 2025

Role: Consultant – Development

Hive • Linux • PySpark • SQL • Decision Trees • K-Means • Linear Regression • Random Forest • Rule-Based Algorithms

- Delivered end-to-end data science solutions for telecom clients in South Africa and Nigeria across the Get-Grow-Keep framework, enabling subscriber segmentation, churn prevention, upsell identification, and next-best-action recommendations.
- Built and deployed models for behavior segmentation, inactivity prediction, upsell opportunities, and profitability analysis.
- Developed next-best-action frameworks to improve subscriber engagement and retention strategies.
- Owned data preparation, feature engineering, model development, deployment, and production support end-to-end.

Monocular Depth Estimation using Deep Learning

Jul 2022 – Dec 2022

Role: Consultant – Development

Python • OpenCV • ResNet-50 • Deep Learning • Computer Vision

- Developed a computer vision solution for monocular depth estimation, predicting pixel-wise depth information from single RGB images using ResNet-50-based architectures.
- Curated, annotated, and augmented large-scale image datasets to improve model training quality and prediction accuracy.
- Built scalable computer vision pipelines supporting depth-aware analytics and vision-driven business applications.

DIGITAMIZE INNOVATIONS PVT. LTD. (TECHFORCE.AI)

Dec 2018 – Jul 2022

Low-Code OCR Training & Document Intelligence Platform

Role: Product Engineer

Python • Django • scikit-learn • MultiOutputRegressor • OpenCV • GCP Storage • AWS S3 • Celery • Docker • REST APIs

- Architected and developed a scalable low-code OCR platform enabling non-technical users to train, retrain, and deploy custom document-extraction models through an intuitive UI.
- Built REST API-based model access frameworks allowing seamless integration of trained OCR models across enterprise applications.
- Engineered document and image classification pipelines for intelligent document categorization and processing automation.
- Implemented asynchronous task processing (Celery) and containerized deployments (Docker) for scalable, consistent environments across AWS S3 and GCP Storage.

AI-Based Signature Verification & Document Authentication System

Role: Product Engineer

YOLOv4 • Siamese Networks • OCR • Python • OpenCV • Docker • Compute Engine • Deep Learning

- Architected an end-to-end AI-powered signature verification and document authentication solution for enterprise account-verification workflows.
- Designed computer vision pipelines using YOLOv4 object detection for handwritten signature localization, and Siamese Networks for similarity-based verification against reference signatures.
- Built OCR-driven validation workflows for account number extraction and cross-verification, reducing verification fraud.
- Developed scalable, containerized inference pipelines (Docker, Compute Engine) for automated, real-time document verification.

EDUCATION

Bachelor of Engineering (Computer Science)

Shivaji University, Kolhapur | 2018